

IIE4124 Optimization Theory and Application

2025-2 Homework 01

Submission Due: 2025.10.02 Thursday 10:00 am

Instruction:

- Homework must be turned in via the submission link posted in LearnUs by each team. Note that homework submitted by one team member will be automatically counted for other team members.
- Late homework submissions will not be accepted unless pre-approved by the instructor with an acceptable reason. Not submitting homework will result in a “0” point.
- Answer sheets can be composed via typing or hand-writing (by using tablet devices or composing answers on paper and scanning them). Answer sheets should be compiled in a single pdf file with a cover page including name and student id attached.
- Show your work explicitly and clearly to get full/partial credit for each problem.
- Grading policy
 - Excellent: 10
 - Very Good: 9
 - Good: 8
 - Satisfactory: 7
 - Not Satisfactory: 6
 - Below than expected: 5
 - No submission: **0**

Reading Assignment:

- Review Chapter 1 and Chapter 2 of the textbook and lecture notes.

Problem Solving Questions

2.6 The questions below walk you through the steps of modeling the selling of an asset (say, a single share of a stock).

- (a) Assume that you are simulating your prices using data from a mathematical model given by

$$p_{t+1} = \eta_0 p_t + \eta_1 p_{t-1} + \epsilon_{t+1} \quad (1)$$

where $\epsilon \sim N(0, 6^2)$. We do not know the value of $\eta = (\eta_0, \eta_1)$, but our belief about the true value of η is that it is multivariate normal, with $\eta \sim MVN(\bar{\eta}, \Sigma)$ where

$$\bar{\eta}_t = \begin{bmatrix} .7 \\ .3 \end{bmatrix}.$$

Assume that the covariance matrix Σ_t is given by

$$\Sigma_t = \begin{bmatrix} (.2)^2 & (.05)^2 \\ (.05)^2 & (.1)^2 \end{bmatrix}.$$

where $\Sigma_{tij} = Cov(\eta_i, \eta_j)$ for $i, j \in (0, 1)$. Assume that at time t , $p_t = 20$, $p_{t-1} = 24$ and we observe $p_{t+1} = 18.2$. Using the equations (19)-(23) in Chapter 2 lecture notes, what is the updated estimates of $\bar{\eta}_{t+1}$ and Σ_{t+1} ? Give the updating equation and compute $\bar{\eta}_{t+1}$ and Σ_{t+1} numerically.

$$\begin{aligned} \phi_{t+1} &= \begin{bmatrix} p_{t+1} \\ p_t \end{bmatrix} \\ \hat{\epsilon}_{t+1} &= \bar{\eta}_0 p_t + \bar{\eta}_1 p_{t-1} - p_{t+1} = 3 \\ \gamma_{t+1} &= 1 + \phi_{t+1}^\top M_t \phi_{t+1} = 20.0696 \\ M_{t+1} &= M_t - \frac{1}{\gamma_{t+1}} (M_t \phi_{t+1} \phi_{t+1}^\top M_t) = \begin{bmatrix} .009841 & -0.007017 \\ -0.007017 & .006997 \end{bmatrix} \\ \bar{\eta}_{t+1} &= \bar{\eta}_t - \frac{1}{\gamma_{t+1}} M_{t+1} \phi_{t+1} \hat{\epsilon}_{t+1} = \begin{bmatrix} .694205 \\ .298172 \end{bmatrix}. \end{aligned}$$

- (b) What is the state variable for this problem? Give the list of variables. Note that you may have to add variables as you progress through the exercise (you do not have all the information at this point). Make sure you include all the information you need to update $\bar{\eta}_{t+1}$ from $\bar{\eta}_t$. How many dimensions does your state variable have (this is the same as asking how many variables are in S_t).

자산의 보유 여부를 나타내는 변수 R_t 을 도입한다.

$$R_t = \begin{cases} 1 & \text{if we are holding the stock at time } t, \\ 0 & \text{if we are no longer holding the stock at time } t. \end{cases}$$

(a)의 풀이에서, $\bar{\eta}_{t+1}$ 을 업데이트 하는 데에 필요한 다음의 변수들 전부가 state variable에 속한다.

$$\begin{aligned} S_t &= (R_t, \phi_t, \bar{\eta}_t, \Sigma_t) \\ &= (R_t, p_t, p_{t-1}, \bar{\eta}_{t0}, \bar{\eta}_{t1}, Var(\eta_i), Var(\eta_j), Cov(\eta_i, \eta_j), Cov(\eta_j, \eta_i)) \end{aligned}$$

따라서 S_t 는 9차원을 갖는다. 이 때, 공분산(Cov)의 대칭성을 고려한다면 S_t 은 8차원으로도 볼 수 있다.

- (c) Our trader makes trading decisions based on a 7-day moving average of prices. Assume we are on day t , and the 7-day moving average is computed using

$$\bar{p}_t = \frac{1}{7} \sum_{t'=t-7+1}^t p_{t'} \quad (2)$$

The trader will sell an asset if $p_t < \bar{p}_t - \theta^{sell}$. Write out the decision rules as a policy $X^\pi(S_t | \theta^{sell})$ that returns 1 if we sell the asset and 0 otherwise.

$$X^\pi(S_t | \theta^{sell}) = \begin{cases} 1 & \text{If } p_t < \bar{p}_t - \theta^{sell} \text{ and } R_t = 1, \forall t \geq 7 \\ 1 & \text{If } t = T \text{ and } R_t = 1 \\ 0 & \text{Otherwise} \end{cases}$$

(d) What is the exogenous information?

ε_{t+1} 계산에 p_{t+1} 이 사용되기에, $t + 1$ 시점의 가격이 exogenous information로 정의된다.

$$W_{t+1} = p_{t+1}$$

(e) Write out the transition equations. You need an equation for each element of S_t .

$$\begin{aligned} S_{t+1} &= S^M(S_t, x_t, W_{t+1}) \\ &= S^M(R_t, \phi_t, \bar{\eta}_t, \Sigma_t, X^\pi(S_t | \theta^{sell}), p_{t+1}) \\ &= \begin{pmatrix} R_{t+1} = R_t \cdot (1 - X^\pi(S_t | \theta^{sell})) \\ \phi_{t+1} = \begin{bmatrix} p_{t+1} \\ p_t \end{bmatrix} \\ \bar{\eta}_{t+1} = \bar{\eta}_t - \frac{1}{\gamma_{t+1}} \Sigma_{t+1} \phi_{t+1} \hat{\varepsilon}_{t+1} \\ \Sigma_{t+1} = \Sigma_t - \frac{1}{\gamma_{t+1}} \Sigma_t \phi_{t+1} \phi_{t+1}^\top \Sigma_t \end{pmatrix} \end{aligned}$$

$$\text{where } \hat{\varepsilon}_{t+1} = \bar{\eta}_t^\top \phi_t - p_{t+1}, \gamma_{t+1} = 1 + \phi_{t+1}^\top \Sigma_t \phi_{t+1}$$

(f) Write out the objective function (and remember to use an expectation operator for each random variable as described in the instructions). Be sure to specify what you are optimizing over given the class of policy specified in part (c). Assume that you are training your policy on historical data in an offline setting.

$$\max_{\pi(\theta)} F^\pi = \mathbb{E} \left[\sum_{t=1}^T p_t X^\pi(S_t) \mid S_1 \right]$$

Historical data를 이용한 sample path들의 집합을 ω^n ($n = 1, \dots, N$) 이라고 할 때, sample path에 대해 policy를 평가하여 실제 성능을 근사시킬 수 있다.

$$\hat{F}^\pi(\omega^n) = \sum_{t=1}^T p_t(\omega^n) X^\pi(S_t(\omega^n)).$$

$$F^\pi \approx \bar{F}^\pi = \frac{1}{N} \sum_{n=1}^N \hat{F}^\pi(\omega^n) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T p_t(\omega^n) X^\pi(S_t(\omega^n)).$$

따라서 sample path를 이용하여 근사한 목적식은 아래와 같다.

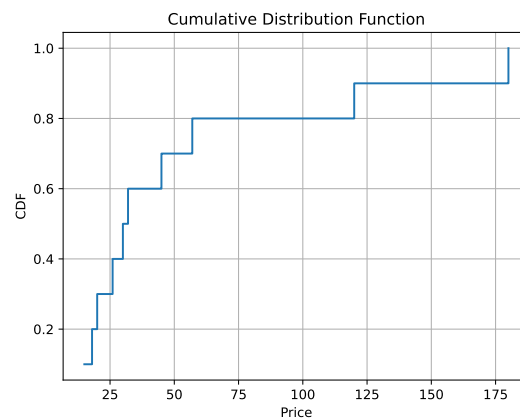
$$\max_{\pi(\theta)} \bar{F}^\pi$$

2.7 You need to run a simulation of electricity prices, which are notoriously heavy-tailed. You collect the data shown in Table 1.

Table 1: Set of prices for 2.7

Time	Price
1	20
2	32
3	26
4	180
5	30
6	45
7	18
8	120
9	57
10	15

- (a) Use the data in the table to produce a cumulative distribution function. You will need to plot the cdf, which looks like a step function.



- (b) You now observe three realizations of prices: 25, 18, 160. Use the cumulative distribution function from (a) to create three realizations of a random variable that is uniformly distributed between 0 and 1 (call this random variable U).

$$U = [0.3, 0.2, 0.9]$$

- (c) Now use these observations of U to create three observations of a random variable Z that is normally distributed with mean 0, variance 1. Be sure that your method for generating these random variables is clear. [Hint: use the cumulative distribution for a normal (0,1) random variable just as you used the cdf you created in part (a) to create uniform random variables in part (b).]

$$\text{Norm.inv}(0.3, 0, 1) = -0.5244005127080409$$

$$\text{Norm.inv}(0.2, 0, 1) = -0.8416212335729142$$

$$\text{Norm.inv}(0.9, 0, 1) = 1.2815515655446004$$

$$Z \approx [-0.5244, -0.8416, 1.2816]$$

- (d) Assume that you used this method for creating a sequence of normal (0,1) random variables to fit a linear model of the form

$$Z_{t+1} = 0.7Z_t + 0.3Z_{t-1} + \varepsilon_{t+1}$$

where ε_{t+1} is normally distributed with mean 0 and variance 1 (we know this because we are simulating normal (0,1) random variables). Starting with $t = 1$ and $Z_0 = 0.5$ and $Z_1 = -0.3$, generate Z_2, Z_3, Z_4 assuming that $\varepsilon_2 = -0.6, \varepsilon_3 = 2.2, \varepsilon_4 = 1.4$. Then, use your cumulative distribution from part (a) to generate observations of P_2, P_3 , and P_4 .

$$Z_2 = 0.7 \cdot (-0.3) + 0.3 \cdot (0.5) - 0.6 = -0.66$$

$$Z_3 = 0.7 \cdot (-0.66) + 0.3 \cdot (-0.3) + 2.2 = 1.648$$

$$Z_4 = 0.7 \cdot (1.648) + 0.3 \cdot (-0.66) + 1.4 = 2.3556$$

위 $[Z_2, Z_3, Z_4]$ 를 표준정규분포의 누적확률로 환산하면 대략 $[0.2546, 0.9503, 0.9908]$ 이 되고, 이를 (a)의 누적확률분포에 대응시켜 값을 생성하면 다음과 같다.

$$P_2 = 20, P_3 = 180, P_4 = 180$$

Programming Questions

These exercises use the Python module *AssetSelling*. Consider a policy that understands that the price of the asset might be rising, which means that static buy-sell limits might not be effective. Assume that we forecast the price for time $t + 1$ using the fitted time series model

$$\bar{p}_t = 0.7p_t + 0.2p_{t-1} + 0.1p_{t-2}.$$

We are going to use $\bar{p}_t$ as a forecast of p_{t+1} given what we know at time t .

- (a) What is the state variable for this problem? If prices are discretized to the nearest 0.1, and prices are assumed to range between 0 and 100, what is the size of the state space?
- (b) Assume we introduce the policy

$$X^{time-series}(S_t|\theta) = \begin{cases} 1 & \text{If } p_t < \bar{p}_t + \theta \text{ or } p_t > \bar{p}_t + \theta, \\ 1 & \text{If } t = T, \\ 0 & \text{Otherwise} \end{cases}. \quad (3)$$

In this version of the policy, we are looking for sudden deviations from the expected price. Plot the graph of the objective function (contribution) vs. θ . Comment on your plot. Note that while the state space is quite large, there is no change in the complexity of finding the best policy in this class.

- (c) Imagine that our stock follows a seasonal pattern, suggesting that our buy-sell signals should be time dependent. This means that we need to replace θ with θ_t . Discuss how this would complicate our policy search process.